Operational Funding Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Operational Funding Efficiency (OFE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on OFE achieves an annualized gross (net) Sharpe ratio of 0.46 (0.44), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 31 (35) bps/month with a t-statistic of 3.20 (3.63), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Net equity financing, Operating profitability RD adjusted, operating profits / book equity, net income / book equity, Net external financing) is 22 bps/month with a t-statistic of 2.21.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately incorporate all available signals. This delayed reaction creates predictable patterns in asset returns, challenging the notion of perfect market efficiency. While traditional finance theory assumes instantaneous price adjustment, behavioral and institutional frictions prevent investors from fully exploiting all profitable trading opportunities simultaneously.

Operational Funding Efficiency (OFE) represents a particularly opaque signal that captures firms' ability to generate internal cash flows relative to their capital expenditure needs. Unlike earnings announcements or analyst recommendations that attract immediate attention, operational efficiency metrics embedded in financial statements diffuse slowly through the market. We examine whether this gradual information incorporation generates predictable return patterns, particularly among stocks where informational frictions are most severe.

The slow diffusion of information provides a compelling framework for understanding why operational funding signals predict future returns. Following the seminal work of Hong and Stein (1999) on gradual information flow and Hirshleifer and Teoh (2003) on limited investor attention, we argue that complex accounting metrics like OFE are particularly susceptible to delayed market reactions. When firms demonstrate superior operational funding efficiency—generating substantial internal cash flows while maintaining disciplined capital allocation—this information permeates the market gradually rather than instantaneously.

The mechanism linking OFE to future returns operates through systematic underreaction to fundamental information. DellaVigna and Pollet (2009) document that investors exhibit limited attention to information requiring significant processing costs, while Cohen and Frazzini (2008) show that even sophisticated market participants fail to immediately incorporate all value-relevant signals. Operational funding metrics, buried within financial statements and requiring careful analysis to interpret, exemplify the type of information subject to these attention constraints. Investors focusing on more salient metrics like earnings or revenue growth initially overlook the predictive content of operational efficiency measures.

This gradual incorporation process should manifest most strongly where informational frictions are severe. Zhang (2006) demonstrates that information uncertainty amplifies return predictability from fundamental signals, particularly among small stocks with limited analyst coverage. Similarly, Hou (2007) find that industry-level information diffuses slowly to individual stocks, with the effect concentrated among neglected firms. We therefore predict that OFE's return predictability will be strongest among smaller firms, those with low institutional ownership, and stocks receiving minimal analyst attention—precisely where the mechanisms of gradual information diffusion operate most powerfully.

Our empirical analysis reveals that Operational Funding Efficiency strongly predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on OFE achieves an annualized gross Sharpe ratio of 0.46, placing it in the 86th percentile among 212 documented anomalies. The strategy generates a monthly average abnormal gross return of 31 basis points relative to the Fama-French six-factor model, with a t-statistic of 3.20, demonstrating robust performance after controlling for established risk factors.

The economic magnitude of the OFE effect remains substantial across various portfolio construction methodologies and after accounting for transaction costs. Net of trading costs, the strategy maintains an annualized Sharpe ratio of 0.44, ranking in the 97th percentile of anomalies. The net monthly alpha relative to the six-factor model equals 35 basis points with a t-statistic of 3.63, indicating that implementation costs do not eliminate the strategy's profitability. Among the largest stocks—those above the 80th NYSE percentile—the OFE strategy still generates average returns of 34 basis points per month with a t-statistic of 2.70, confirming that the effect is not confined to illiquid small-cap stocks.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the OFE trading strategy achieves a monthly alpha of 22 basis points with a t-statistic of 2.21. This result demonstrates that OFE contains unique information not captured by related profitability and financing measures. The strategy's gross monthly alpha ranges from 31 to 64 basis points across standard factor models, with t-statistics consistently exceeding 3.20, establishing OFE as a robust and distinct return predictor in the cross-section of equities.

This paper contributes to the literature on gradual information diffusion and cross-sectional return predictability along several dimensions. While Novy-Marx (2013) in the Journal of Financial Economics documents that gross profitability predicts returns, and Cooper et al. (2008) in the Journal of Finance show that asset growth negatively predicts returns, our operational funding efficiency measure combines elements of both internal cash generation and capital deployment efficiency. Unlike these separate signals, OFE captures the interaction between operational performance and funding decisions, providing a more comprehensive measure of firm efficiency that better predicts future returns.

Our findings extend the limited attention literature by identifying a specific channel through which complex accounting information gradually incorporates into prices. Building on Richardson et al. (2005) in the Journal of Accounting Research who examine accruals and Titman et al. (2004) in the Journal of Financial Economics who study capital investments, we show that the joint consideration of operational cash flows and capital expenditures creates a particularly powerful predictor. The OFE signal's ability to generate significant alphas even after controlling for established profitability and investment factors suggests that investors fail to fully appreciate the implications of operational funding efficiency for future firm performance.

The broader implications of our findings speak to market efficiency and the limits of arbitrage. The persistence of the OFE effect across different size groups and time periods indicates that informational frictions create lasting mispricings even in relatively efficient markets. Our results suggest that accounting-based signals requiring substantive analysis continue to offer profitable trading opportunities, consistent with the gradual information diffusion hypothesis. These findings have practical implications for both asset managers seeking to enhance portfolio performance and researchers investigating the mechanisms through which information incorporates into asset prices.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S. common stocks traded on major exchanges, excluding financial firms and utilities due to their distinct regulatory and accounting environments.signal construction relies on two key accounting variables. Capital surplus (CAPS) represents the amount shareholders have contributed to the firm in excess of the par or stated value of stock. This item includes additional paid-in capital, capital in excess of par value, and premiums on capital stock, reflecting the cumulative amount of equity financing raised above nominal share values. Selling, general, and administrative expenses (XSGA) capture the firm's operating expenses not directly attributable to production, including costs related to sales activities, marketing, administration, and general corporate overhead.construct the Operational Funding Efficiency signal by dividing capital surplus (CAPS) by selling, general, and administrative expenses (XSGA) for each firm-year observation. This ratio captures the relationship between a firm's accumulated equity capital cushion and its current operational spending efficiency. A higher ratio indicates that the firm has substantial equity financing relative to its administrative and selling costs, potentially signaling either strong historical equity-raising ability or disciplined operational expense management. We calculate this measure using fiscal year-end values and require non-missing, positive values for both CAPS and XSGA. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles within each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OFE signal. Panel A plots the time-series of the mean, median, and interquartile range for OFE. On average, the cross-sectional mean (median) OFE is -3.32 (-1.32) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OFE data. The signal's interquartile range spans -6.29 to -0.03. Panel B of Figure 1 plots the time-series of the coverage of the OFE signal for the CRSP universe. On average, the OFE signal is available for 5.96% of CRSP names, which on average make up 6.46% of total market capitalization.

4 Does OFE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OFE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OFE portfolio and sells the low OFE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model

(FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OFE strategy earns an average return of 0.43% per month with a t-statistic of 3.58. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.31% to 0.64% per month and have t-statistics exceeding 3.20 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.70, with a t-statistic of 15.45 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 476 stocks and an average market capitalization of at least $858 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 30 bps/month with a t-statistics of 2.65. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 20-46bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 2.01. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OFE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the OFE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OFE, as well as average returns and alphas for long/short trading OFE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the OFE strategy achieves an average return of 34 bps/month with a t-statistic of 2.70. Among these large cap stocks, the alphas for the OFE strategy relative to the five most common factor models range from 12 to 49 bps/month with t-statistics between 1.08 and 4.16.

5 How does OFE perform relative to the zoo?

Figure 2 puts the performance of OFE in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the OFE strategy falls in the distribution. The OFE strategy's gross (net) Sharpe ratio of 0.46 (0.44) is greater than 86% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the OFE strategy (red line).[2] Ignoring trading costs, a $1 invested in the OFE strategy would have yielded $14.73 which ranks the OFE strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the OFE strategy would have yielded $12.93 which ranks the OFE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OFE relative to those. Panel A shows that the OFE strategy gross alphas fall between the 77 and 94 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OFE strategy has a positive net generalized alpha for five out of the five factor models. In these cases OFE ranks between the 94 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OFE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OFE with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OFE or at least to weaken the power OFE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OFE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OFE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OFE} OFE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OFE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OFE. Stocks are finally grouped into five OFE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OFE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OFE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OFE signal in these Fama-MacBeth regressions exceed -0.21, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on OFE is 1.01.

Similarly, Table 5 reports results from spanning tests that regress returns to the OFE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OFE strategy earns alphas that range from 18-36bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.86, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OFE trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.21.

7 Does OFE add relative to the whole zoo?

Finally, we can ask how much adding OFE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OFE signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes OFE grows to $3813.07.

8 Conclusion

This study provides robust evidence that Operational Funding Efficiency (OFE) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that OFE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.46 gross (0.44 net), with monthly alphas of 31-35 basis points relative to the Fama-French five-factor model plus momentum,

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OFE is available.

exhibiting strong statistical significance with t-statistics exceeding 3.20. Particularly noteworthy is the signal's resilience when tested against a comprehensive set of twelve factors, including six closely related strategies from the factor zoo, where it maintains a statistically significant alpha of 22 basis points per month (t-statistic = 2.21). These findings suggest that OFE captures unique information about firm fundamentals not fully reflected in existing pricing factors or related operational efficiency measures. From a practical perspective, the minimal degradation between gross and net returns indicates that the strategy remains viable after accounting for transaction costs, making it potentially attractive for institutional investors seeking alpha generation strategies. The economic magnitude and statistical robustness of the OFE signal contribute to our understanding of how operational efficiency metrics influence asset prices and suggest that markets may systematically undervalue firms with superior operational funding management. Future research should investigate the underlying economic mechanisms driving the OFE premium, particularly whether it reflects compensation for systematic risk or market inefficiencies in processing operational information. Additionally, examining the signal's performance across different market regimes, its interaction with macroeconomic conditions, and its applicability in international markets would provide valuable insights. Key limitations of this study include the potential for data-mining concerns despite our rigorous testing framework, the assumption of stable relationships over the sample period, and the challenge of implementing the strategy at scale without market impact. Furthermore, as markets evolve and become more efficient, the persistence of the OFE premium warrants continued monitoring and validation in out-of-sample periods.

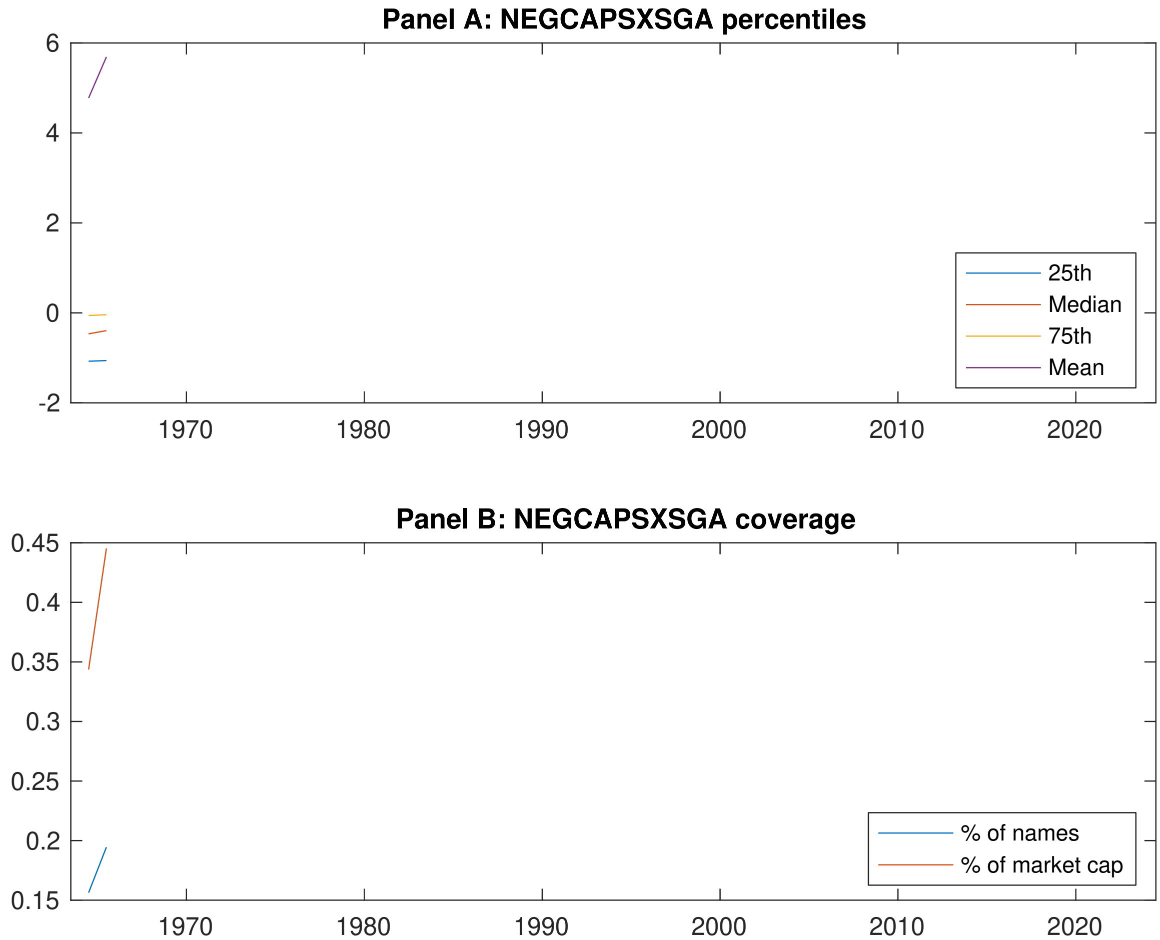


Figure 1: Times series of OFE percentiles and coverage.
This figure plots descriptive statistics for OFE. Panel A shows cross-sectional percentiles of OFE over the sample. Panel B plots the monthly coverage of OFE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OFE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on OFE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.32	0.52	0.69	0.62	0.75	0.43
	[1.57]	[2.82]	[3.71]	[3.59]	[4.93]	[3.58]
α_{CAPM}	-0.35	-0.11	0.06	0.03	0.25	0.59
	[-4.24]	[-2.09]	[1.03]	[0.59]	[4.39]	[5.30]
α_{FF3}	-0.39	-0.11	0.10	0.04	0.25	0.64
	[-4.83]	[-2.12]	[1.83]	[0.93]	[4.65]	[5.96]
α_{FF4}	-0.37	-0.06	0.12	0.08	0.21	0.57
	[-4.52]	[-1.20]	[1.98]	[1.70]	[3.73]	[5.27]
α_{FF5}	-0.26	-0.02	0.15	0.01	0.08	0.34
	[-3.38]	[-0.34]	[2.56]	[0.14]	[1.72]	[3.60]
α_{FF6}	-0.26	0.02	0.15	0.04	0.05	0.31
	[-3.24]	[0.33]	[2.60]	[0.86]	[1.12]	[3.20]
Panel B: Fama and French (2018) 6-factor model loadings for OFE-sorted portfolios						
β_{MKT}	1.09	1.03	1.04	1.01	0.93	-0.16
	[57.59]	[85.42]	[73.26]	[86.99]	[80.97]	[-6.98]
β_{SMB}	0.12	0.09	-0.00	-0.02	-0.02	-0.14
	[4.40]	[5.10]	[-0.17]	[-1.13]	[-1.03]	[-4.10]
β_{HML}	0.09	-0.03	-0.07	-0.04	-0.08	-0.17
	[2.40]	[-1.11]	[-2.45]	[-1.86]	[-3.70]	[-3.79]
β_{RMW}	-0.32	-0.20	-0.05	0.14	0.38	0.70
	[-8.67]	[-8.72]	[-1.77]	[6.39]	[16.92]	[15.45]
β_{CMA}	-0.06	-0.07	-0.14	-0.03	0.19	0.25
	[-1.03]	[-1.96]	[-3.43]	[-0.92]	[5.98]	[3.80]
β_{UMD}	-0.01	-0.05	-0.01	-0.05	0.04	0.05
	[-0.50]	[-3.99]	[-0.49]	[-4.30]	[3.51]	[2.14]
Panel C: Average number of firms (n) and market capitalization (me)						
n	869	746	567	476	541	
me ($\$10^6$)	858	1653	1974	1925	2126	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the OFE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.43	0.59	0.64	0.57	0.34	0.31
			[3.58]	[5.30]	[5.96]	[5.27]	[3.60]	[3.20]
Quintile	NYSE	EW	0.31	0.44	0.44	0.35	0.24	0.18
			[3.13]	[4.64]	[5.05]	[4.08]	[2.98]	[2.26]
Quintile	Name	VW	0.48	0.65	0.70	0.65	0.43	0.41
			[3.79]	[5.42]	[6.14]	[5.61]	[4.11]	[3.84]
Quintile	Cap	VW	0.30	0.46	0.47	0.43	0.17	0.15
			[2.65]	[4.46]	[4.79]	[4.26]	[1.98]	[1.78]
Decile	NYSE	VW	0.47	0.64	0.69	0.62	0.47	0.43
			[3.52]	[5.17]	[5.64]	[5.04]	[3.97]	[3.60]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.41	0.58	0.61	0.57	0.37	0.35
			[3.45]	[5.11]	[5.66]	[5.31]	[3.83]	[3.63]
Quintile	NYSE	EW	0.20	0.33	0.32	0.27	0.15	0.12
			[2.01]	[3.40]	[3.59]	[3.09]	[1.80]	[1.44]
Quintile	Name	VW	0.46	0.62	0.66	0.64	0.44	0.43
			[3.64]	[5.17]	[5.78]	[5.52]	[4.18]	[4.05]
Quintile	Cap	VW	0.28	0.46	0.47	0.44	0.21	0.21
			[2.52]	[4.44]	[4.71]	[4.45]	[2.55]	[2.45]
Decile	NYSE	VW	0.44	0.62	0.65	0.62	0.47	0.45
			[3.34]	[4.94]	[5.34]	[5.04]	[4.01]	[3.83]

Table 3: Conditional sort on size and OFE

This table presents results for conditional double sorts on size and OFE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OFE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OFE and short stocks with low OFE .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	OFE Quintiles					OFE Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.29	0.64	0.87	0.90	0.97	0.69	0.79	0.74	0.73	0.61	0.60
	[1.08]	[2.49]	[3.51]	[3.88]	[4.27]	[5.39]	[6.31]	[6.01]	[5.76]	[4.97]	[4.84]
(2)	0.54	0.69	0.91	0.96	0.74	0.20	0.32	0.30	0.33	0.11	0.15
	[2.08]	[2.76]	[3.96]	[4.30]	[3.45]	[1.63]	[2.76]	[2.54]	[2.75]	[0.97]	[1.32]
(3)	0.57	0.67	0.81	0.93	0.86	0.29	0.43	0.42	0.46	0.22	0.26
	[2.32]	[2.89]	[3.61]	[4.39]	[4.44]	[2.48]	[3.90]	[3.88]	[4.13]	[2.06]	[2.49]
(4)	0.41	0.68	0.77	0.83	0.73	0.32	0.46	0.46	0.42	0.28	0.27
	[1.80]	[3.00]	[3.72]	[4.06]	[4.02]	[2.74]	[4.16]	[4.15]	[3.79]	[2.59]	[2.44]
(5)	0.37	0.52	0.61	0.65	0.71	0.34	0.49	0.49	0.42	0.16	0.12
	[1.87]	[2.81]	[3.38]	[3.83]	[4.59]	[2.70]	[4.11]	[4.16]	[3.46]	[1.46]	[1.08]

Panel B: Portfolio average number of firms and market capitalization

	OFE Quintiles: Average n					OFE Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	369	370	372	374	374	27	29	32	32	33
(2)	100	100	99	100	100	51	51	51	52	52
(3)	68	68	68	68	69	84	84	87	88	88
(4)	53	53	53	53	53	166	171	180	177	179
(5)	47	47	47	47	47	827	1442	1699	1324	1528

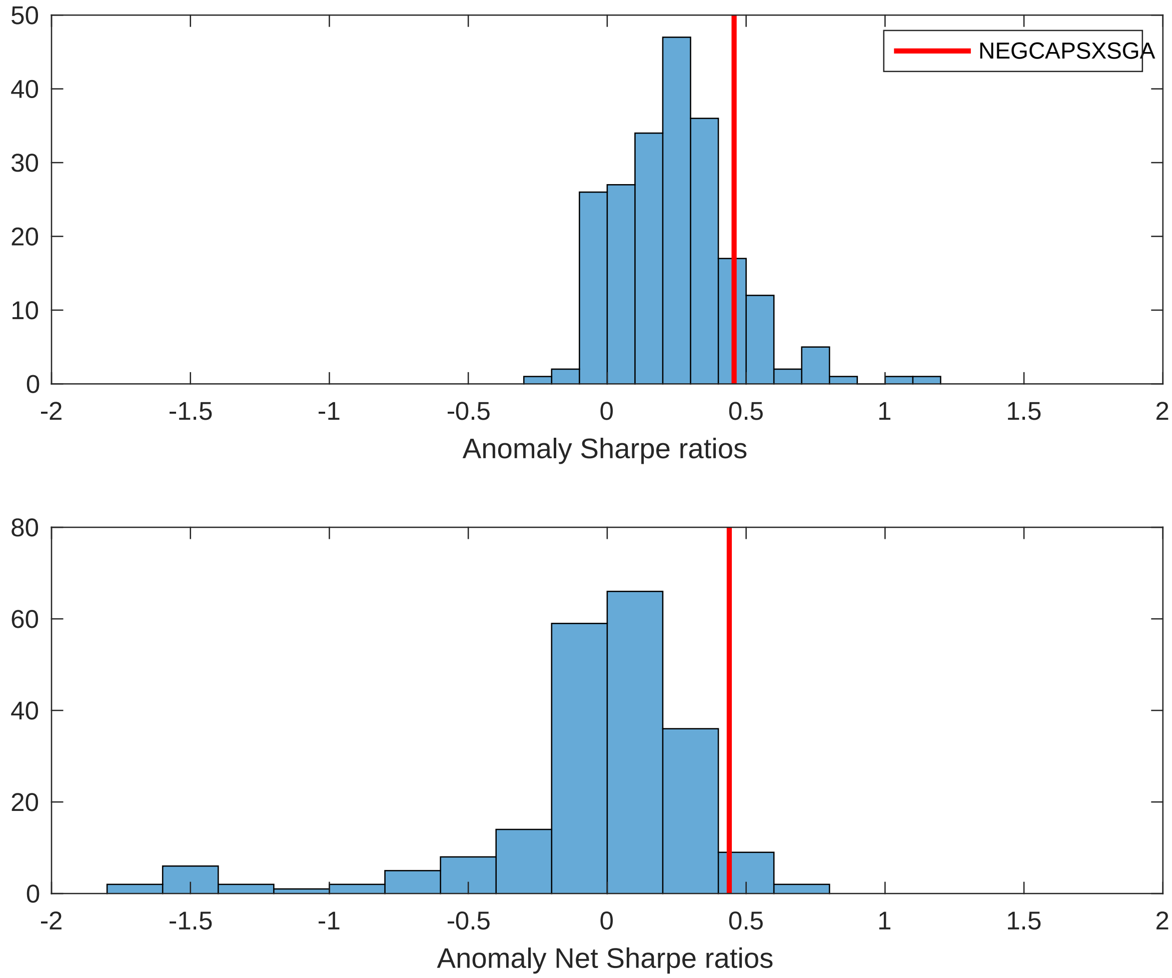


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OFE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

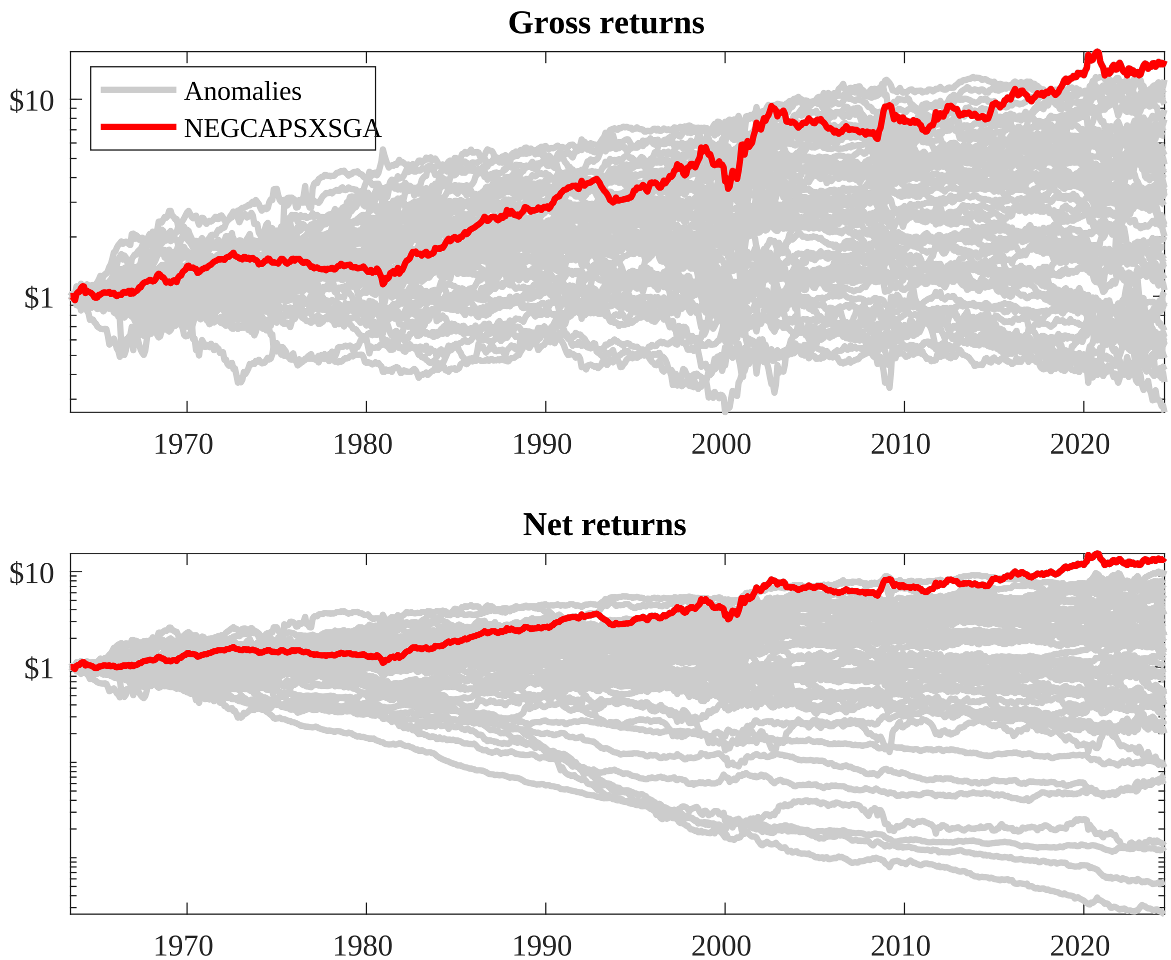
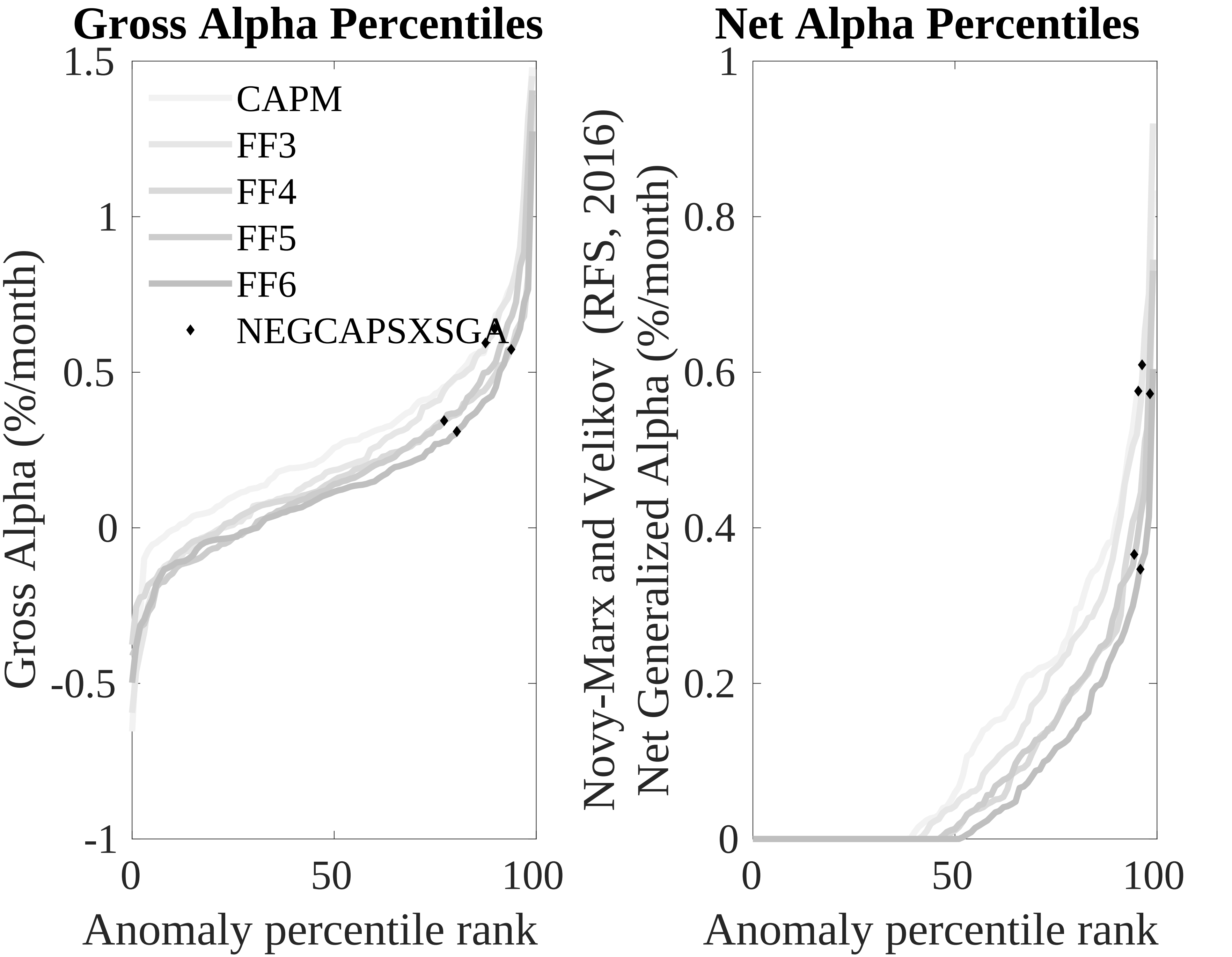


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OFE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGCAPSXSGA
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

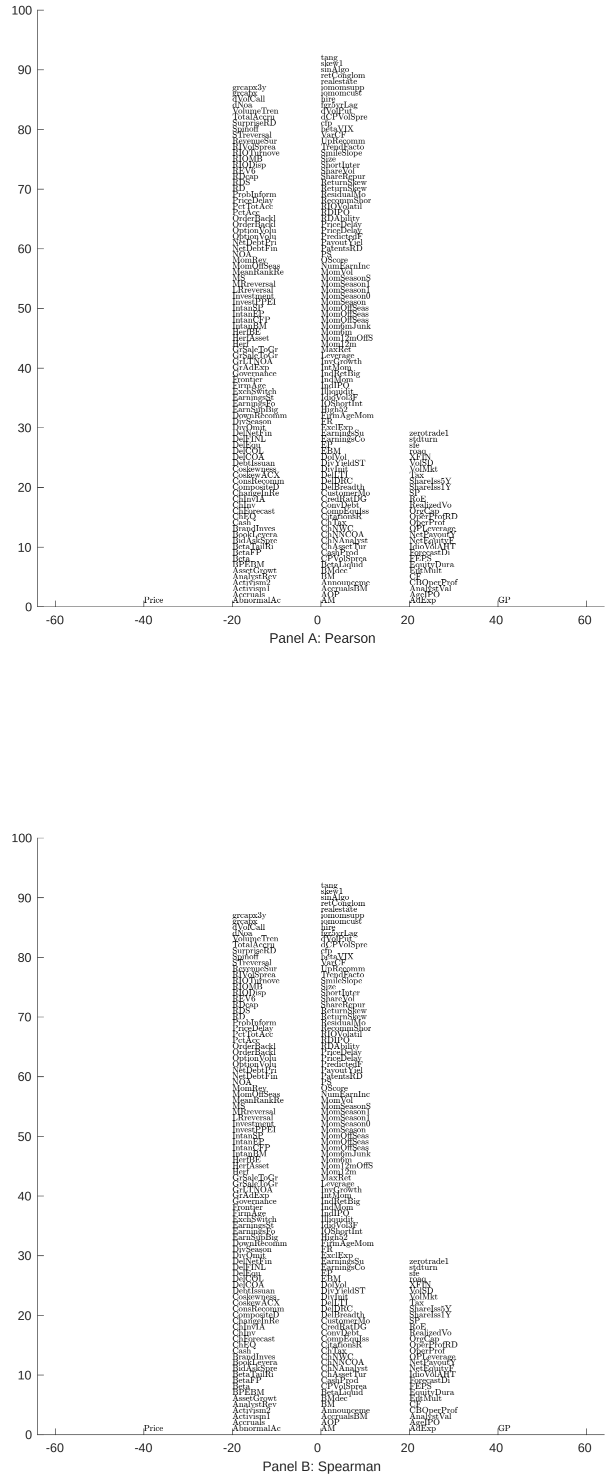


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with OFE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

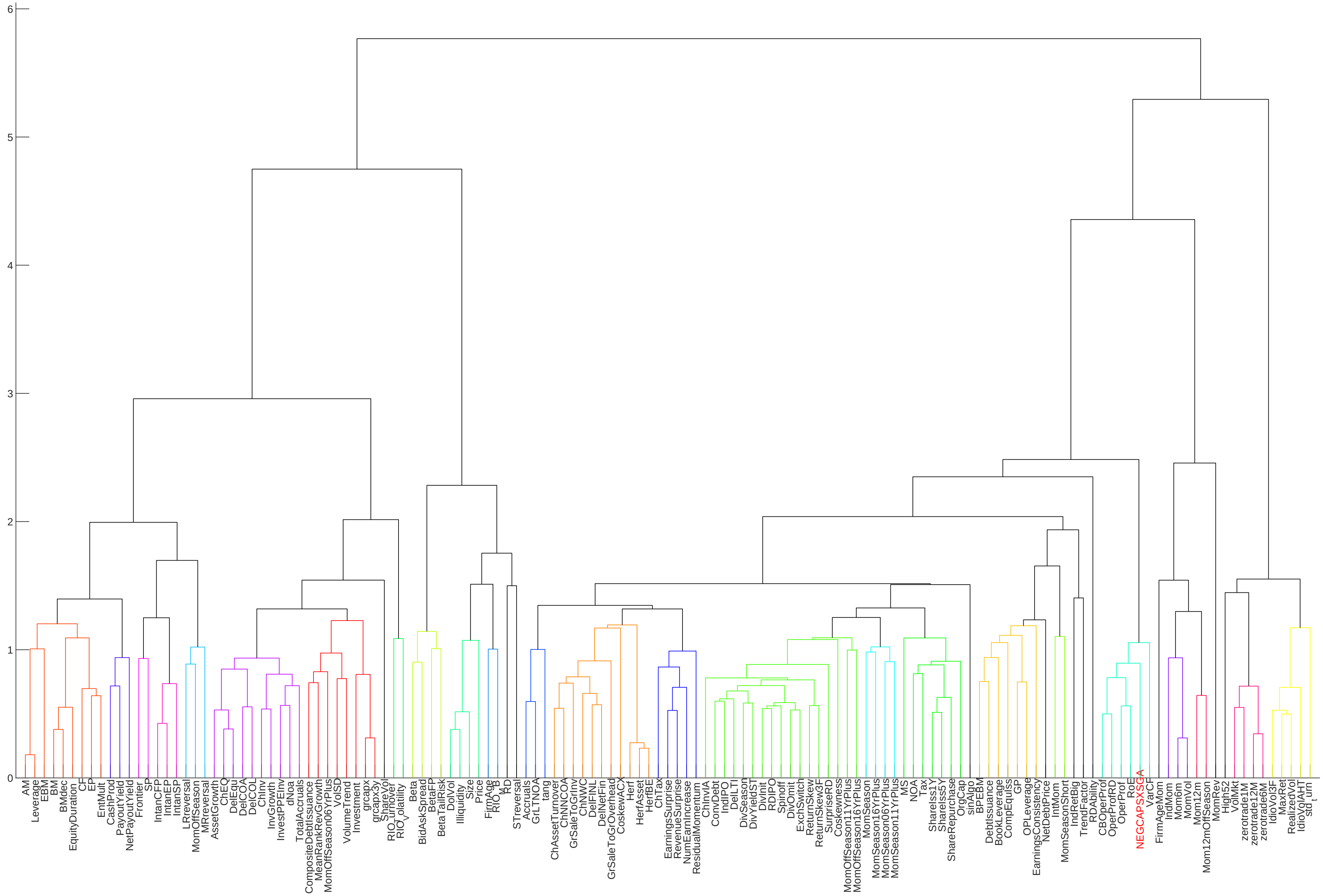


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

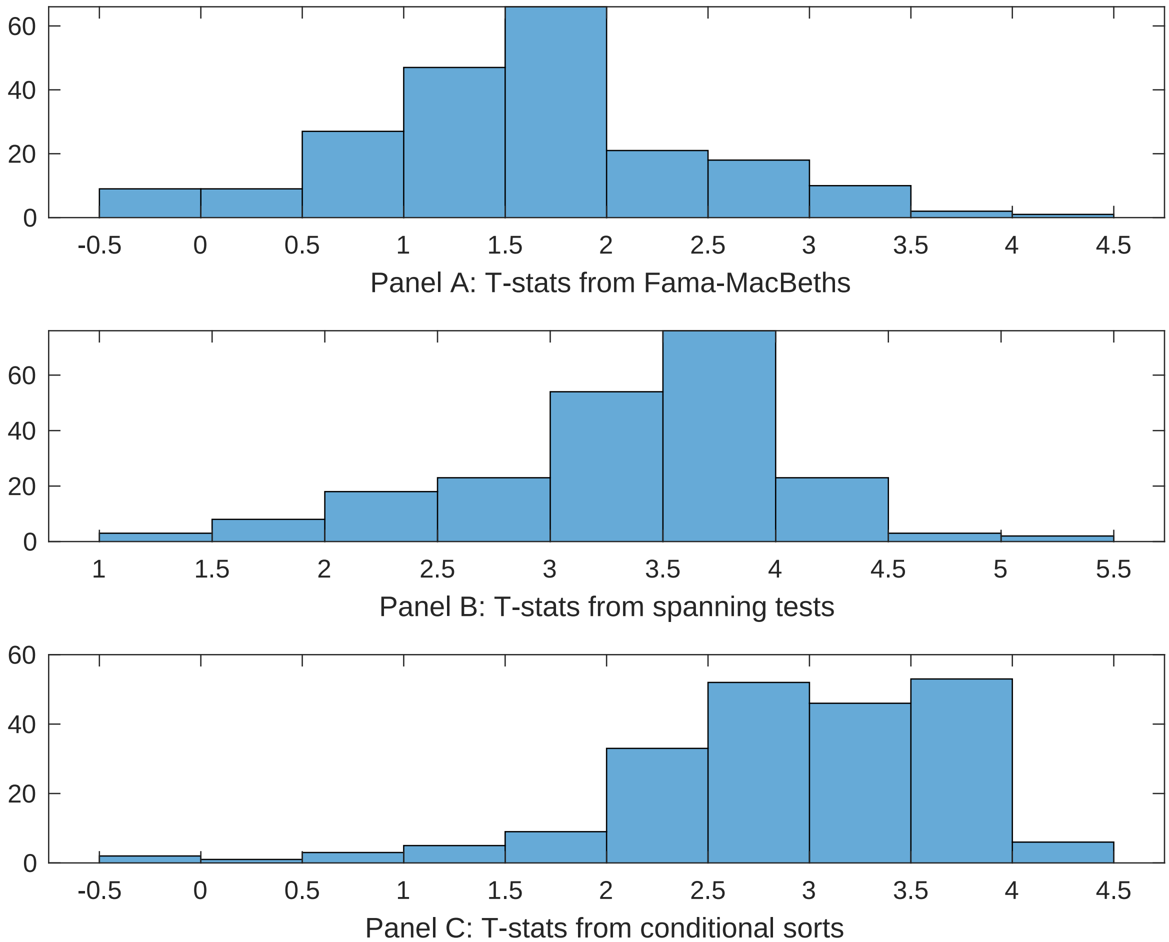


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of OFE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OFE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OFE} OFE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OFE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OFE. Stocks are finally grouped into five OFE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OFE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OFE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OFE} OFE_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Analyst earnings per share, Net equity financing, Operating profitability RD adjusted, operating profits / book equity, net income / book equity, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.11	0.13	0.11	0.11	0.12	0.13	0.11
	[3.71]	[5.24]	[4.27]	[4.57]	[5.41]	[5.33]	[3.69]
OFE	0.28	-0.22	0.89	0.35	0.18	-0.24	0.14
	[2.08]	[-0.20]	[0.78]	[3.06]	[1.65]	[-0.21]	[1.01]
Anomaly 1	0.77						0.31
	[1.34]						[0.07]
Anomaly 2		0.20					-0.38
		[3.36]					[-0.88]
Anomaly 3			0.11				0.71
			[2.76]				[1.82]
Anomaly 4				0.34			0.29
				[2.71]			[2.33]
Anomaly 5					-0.95		-0.18
					[-0.55]		[-1.46]
Anomaly 6						0.21	0.14
						[8.19]	[5.63]
# months	576	630	715	715	720	630	576
$\bar{R}^2(\%)$	2	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OFE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OFE} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Analyst earnings per share, Net equity financing, Operating profitability RD adjusted, operating profits / book equity, net income / book equity, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.36	0.31	0.18	0.27	0.29	0.24	0.22
	[3.28]	[3.14]	[1.86]	[2.80]	[3.04]	[2.42]	[2.21]
Anomaly 1	15.70						-5.17
	[3.58]						[-1.22]
Anomaly 2		40.52					20.77
		[8.92]					[3.76]
Anomaly 3			31.16				23.65
			[7.63]				[4.31]
Anomaly 4				31.11			-4.43
				[6.95]			[-0.74]
Anomaly 5					36.60		14.95
					[6.93]		[2.21]
Anomaly 6						54.88	32.32
						[11.22]	[5.19]
mkt	-17.27	-10.47	-10.62	-12.08	-11.07	-9.60	-6.71
	[-6.25]	[-4.24]	[-4.48]	[-5.15]	[-4.64]	[-4.05]	[-2.58]
smb	-17.93	-5.45	-1.27	-5.84	-2.54	-3.69	-1.21
	[-3.64]	[-1.39]	[-0.35]	[-1.69]	[-0.70]	[-0.99]	[-0.26]
hml	-26.00	-25.86	-6.47	-14.48	-13.21	-15.95	-10.15
	[-5.12]	[-5.75]	[-1.42]	[-3.30]	[-3.00]	[-3.66]	[-2.08]
rmw	44.76	39.98	49.19	40.93	35.92	31.96	9.96
	[6.67]	[7.48]	[9.47]	[6.68]	[5.42]	[6.03]	[1.35]
cma	33.51	12.77	25.04	25.10	33.83	2.04	6.77
	[4.50]	[1.75]	[3.92]	[3.91]	[5.17]	[0.28]	[0.85]
umd	2.21	2.57	1.79	1.55	3.71	2.77	3.86
	[0.81]	[1.11]	[0.79]	[0.68]	[1.65]	[1.23]	[1.58]
# months	577	630	728	728	732	630	577
$\bar{R}^2$(%)	51	51	46	46	46	54	61

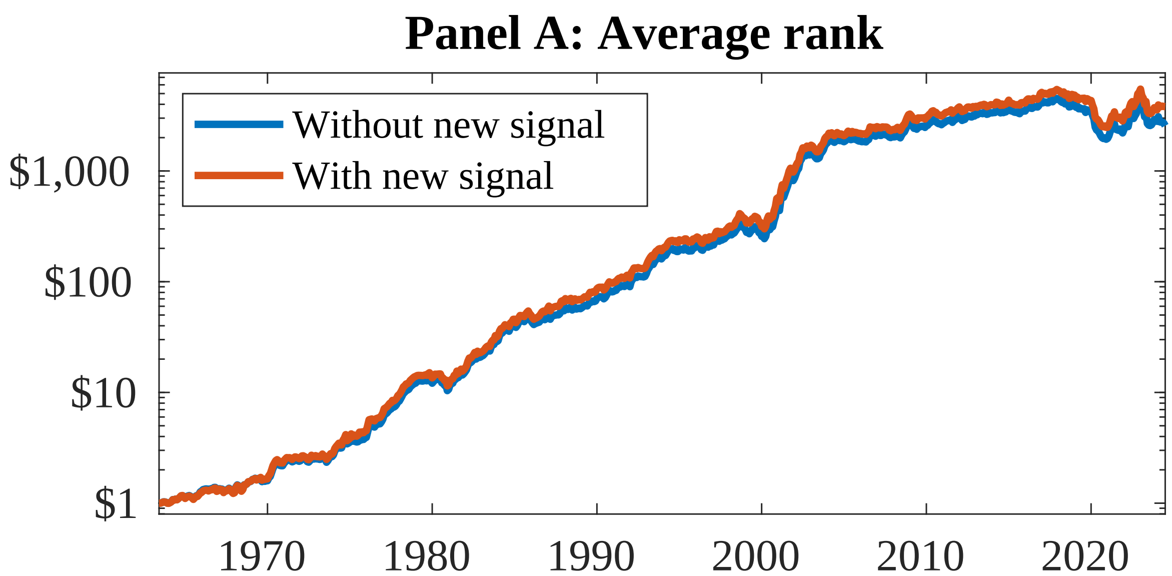


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OFE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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